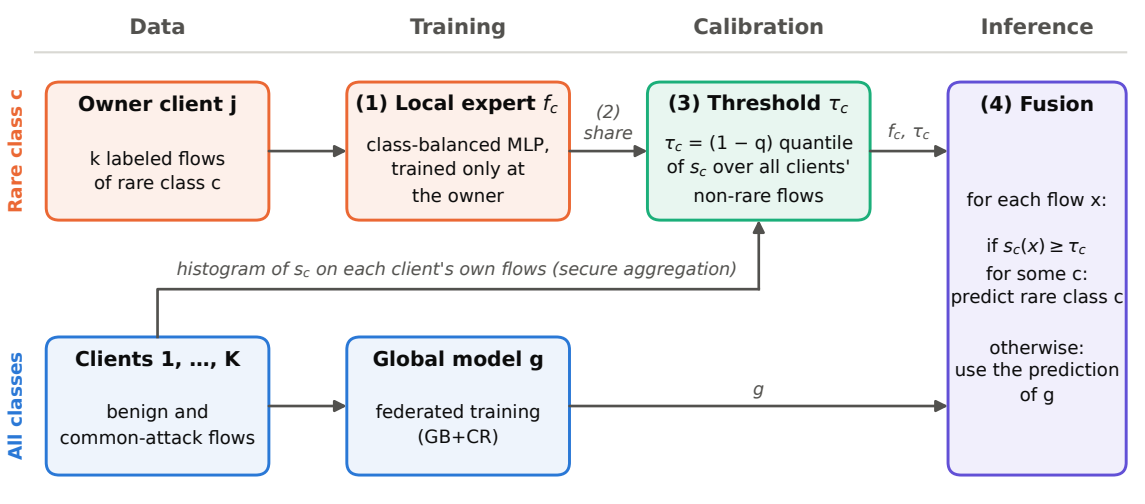
Data
Training
Calibration
Inference
Rare class c
All classes
Owner client j
k labeled flows of rare class c
(1) Local expert f_c
class-balanced MLP, trained only at the owner
(2) share
(3) Threshold τ_c
τ_c = (1 − q) quantile of s_c over all clients' non-rare flows
f_c, τ_c
(4) Fusion
for each flow x:
if $s_c(x) \geq \tau_c$ for some c: predict rare class c
otherwise: use the prediction of g
histogram of s_c on each client's own flows (secure aggregation)
Clients 1, …, K
benign and common-attack flows
Global model g
federated training (GB+CR)
g